

Phase 3 Design Addendum

Decisions from April 29, 2026 design session

This document captures new and updated design decisions made on April 29, 2026, in preparation for Phase 3 (session generator algorithm). It should be absorbed into `PRACTICE_SESSIONS_DESIGN_3.md` at the start of Phase 3 build, and the relevant section of `BUILD_SEQUENCER_2.md` updated to reflect Phase 2 Step 7.

1. Algorithm spec updates (corrections to existing Steps 1–9)

Step 3 — Candidate item pool: `getCandidatesForGoal()` translation layer (NEW)

The existing spec says items enter the pool when “goals reference them.” This is insufficient for coverage goals, which reference entire modules or sub-areas rather than specific items.

Add `getCandidatesForGoal(goal)` **as a core primitive called in Step 3.** This helper expands each goal type into a concrete set of itemRefs:

- **Coverage goal (overall)** — pull all `spacingState` rows where `itemRef` matches the module's `itemRef` prefix patterns AND `acquisition_stage` is not yet in `COVERED_STAGES` (`acquired`, `consolidated`, `mastered`). Items with no `spacingState` row yet (never touched) are also valid candidates — they're implicitly `new` stage.
- **Coverage goal (specific sub-area)** — same logic, filtered to the `itemRef` patterns for the declared sub-area only.
- **Accuracy goal** — all items in the relevant module/group, regardless of stage.
- **Consistency goal** — all items in the relevant module (consistency goals boost the module broadly, not specific items).
- **Umbrella goal** — delegate to children; weight applied at child record level, not umbrella level.
- **Yearly anchor** — delegate to dimension children (Breadth/Mastery/Depth/Consistency records). Each dimension child is a real goal record with a metric type the algorithm handles directly.

Step 4 — Goal-alignment weighting: per-goal-type mapping (UPDATED)

The existing spec describes goal-alignment weighting without accounting for the richer goal types now in the system. Updated mapping:

- **Coverage goals** — items returned by `getCandidatesForGoal` get a goal-alignment boost. Boost scales with urgency (see pace-based urgency below).
- **Accuracy goals** — items in the relevant module/group get boosted. Same as originally spec'd.
- **Consistency goals** — every candidate item from the relevant module gets a mild lift. Signal is "practice this module more often," not "practice these specific items."
- **Umbrella goals** — weight distributed to children, not applied at umbrella level directly.
- **Yearly anchor dimension children** — treated as regular goal records of their respective metric type. No special casing required.

Step 6 — Within-day spacing: explicitly out of scope for Phase 3 (CORRECTION)

The algorithm spec includes Step 6 (within-day spacing for acquiring items) as part of the 9-step flow. The phasing roadmap correctly defers this to Phase 6. **Phase 3 skips Step 6 entirely.** Items in `acquiring` stage still surface regularly via the acquisition factor boost in Step 4. The within-day reinforcement logic (checking if an item was already touched today and ensuring a second touch) is a Phase 6 addition.

Phase 3 algorithm steps: **1, 2, 3, 4, 5, 7, 8, 9**. Step 6 is a clearly marked placeholder stub.

2. Pace-based urgency for coverage goals (NEW)

The problem

Urgency weighting is time-pressure-based. Yearly coverage goals have low time pressure in April, so they lose the weighting battle to monthly and weekly goals every session. Without intervention, a yearly coverage goal makes no progress until it becomes urgent — wrong behavior for something requiring sustained breadth work across 12 months.

Solution: pace score calculation

For any coverage goal (weekly, monthly, or yearly), calculate a pace score alongside target-date proximity:

- **Expected coverage** = (days elapsed in goal period / total days in goal period) × total items
- **Actual coverage** = `getCoverageCount(metric)` — items already at acquired/consolidated/mastered
- **Pace deficit** = expected – actual
- **Pace ratio** = actual / expected (below 1.0 = behind, above 1.0 = ahead)

Weight mapping:

- Ahead of pace (ratio > 1.0) → mild ambient boost
- On pace (ratio ~1.0) → moderate boost, steady momentum
- Behind pace (ratio < 1.0) → boost escalates proportionally; can reach monthly-goal-level urgency when significantly behind

Applies to all scopes

- **Weekly goals** — pace is near real-time. If it's Thursday and you've done 1 of 4 planned sessions, the deficit is immediate and the signal is strong and specific.
- **Monthly goals** — same pace logic, compressed window. Mid-month deficit generates a moderate-urgency signal.
- **Yearly goals** — pace ratio drives a steady ambient pull toward uncovered items throughout the year, ensuring coverage goals make meaningful progress rather than being ignored until October.

User-facing surface

Pace deficit surfaces in the **"Why this plan?" reasoning panel only** — not on the goal row itself. Goal row shows raw progress (43/143). Reasoning panel explains why today's session looks the way it does.

Example reasoning panel language:

- "You're 12 items behind pace on your Ear Training yearly coverage goal — pulling in some uncovered intervals and chord shapes."
- "You need 3 ET sessions before Sunday to stay on pace with your weekly goal."

3. Goal feasibility checking (NEW — Phase 2 Step 7)

The problem

Pace-based urgency ensures items are pulled into sessions appropriately, but doesn't answer: "Do I have enough total practice time in my week to actually hit all my goals?" A user starting May 1 with ambitious yearly coverage goals across all modules needs to know whether their declared day profile mix gives them enough time — proactively, not when it's too late.

`getGoalFeasibility()` helper

New helper in `src/modules/goals/progress.ts` alongside existing helpers.

Inputs:

- All active goals across all scopes
- Current coverage counts (`getCoverageCount`)
- Days remaining in each goal period
- User's declared weekly day profile mix (e.g. 3 Standard + 1 Deep + 1 Light)
- Average items coverable per session type (tunable parameters, initial estimates at build time)

Output: Per-goal feasibility status:

- `on_track` — projected outcome meets or exceeds target
- `at_risk` — projected outcome is close but requires consistency
- `infeasible` — at current day profile mix, target is not reachable by deadline
- Projected outcome value (e.g. "projected 280/408 Shapes items by Dec 31")

Applies to: all active goals across all scopes (weekly, monthly, yearly).

Urgency hierarchy for feasibility signals

Feasibility signals are urgency-tiered, same as pace signals:

- **Weekly behind pace** → critical urgency, days-remaining framing, very direct language
- **Monthly behind pace** → moderate urgency, weeks-remaining framing
- **Yearly behind pace** → ambient signal, pace ratio framing, softer language

Goals home surface — diagnostic view

Per-goal feasibility displayed on the yearly anchor row and monthly/weekly goal rows.
Shows projected vs. target. Flags which goals need more practice time.

Examples:

Ear Training 2026 — on pace ✓

Shapes & Patterns 2026 — at current mix, projected Dec coverage: 280/408. Consider more Deep days.

Chord Progressions (this month) — behind pace. Need 2 more sessions this week.

Practice Sessions home surface — action signal banner

A persistent banner on Practice Sessions home when any goal is behind pace. **Not just one goal — all behind-pace goals are represented, with the most urgent leading.**

Banner structure:

- **Primary row** (always visible) — most urgent behind-pace goal with direct action language + [Deep day] shortcut button
- **Collapsed section** (revealed on tap) — all other behind-pace goals listed with scope and urgency level
- Expand/collapse chevron: "3 more goals behind pace ↓"

When nothing is behind pace — banner disappears entirely. No noise when on track.

Language scales with urgency:

- Weekly: "You need 3 ET sessions before Sunday." → [Deep day]
- Monthly: "You're behind pace on Shapes & Patterns this month. Consider a Deep session."
- Yearly: "Two yearly goals are behind pace. Today's a good day to go deep."

Tapping the banner (or the [Deep day] button) pre-selects Deep as the suggested day profile when the input questionnaire opens.

Phasing

- `getGoalFeasibility()` helper + Goals home surface → **Phase 2, Step 7** (new step, builds on top of Goals home redesign in Step 6)
 - Practice Sessions home banner → **Phase 3** (when Practice Sessions home gets real content anyway)
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4. Block visualization design (NEW — April 29, 2026)

Single block — default state

Each block in the session stack shows:

- **Full block background** in the module's color (not just a left accent bar — the entire block is colored)
- **Module name** (top, small, uppercase)
- **Activity description** — adapts per module type:
 - Declarative (ET, HF): activity name + attempt count ("Chord Function cards — 10 attempts")
 - Procedural (Shapes): activity name + duration ("Chord shape drills — 12 min")
 - Integration (Songs): song names + section + keys ("Mirror, Alpha & Omega · Verse · C, G")
 - Production: path name + duration ("Workflow Foundations · 12 min")
- **Duration** (right-aligned, prominent)
- **Warm-up badge** on the first block when applicable

No memory type labels shown — the display adapts per module without naming the type.

Single block — expanded state (tap to reveal)

Tapping a block reveals:

- **Why snippet** — concise reason this block is in the session ("4 overdue cards · chord function goal due in 9 days · last practiced 5 days ago")
- **Quick-launch button** — "↗ Open [Module Name]" — jumps directly into the relevant module
- **Songs only:** specific section + key combination targets shown in the why snippet

Session stack

- Blocks stacked vertically, proportional height to time allocation
- 2px gap between blocks, slight padding and rounding on the stack container
- Typical session: 4–5 blocks for balanced option, 1–2 blocks for focused option

Two proposal cards — primary tension

The two options always represent **breadth vs. depth**:

- **Card 1 (default)**: balanced, 4–5 blocks, broad module coverage — “Stay on track overall”
- **Card 2**: focused, 1–2 blocks, depth on the highest-priority item — “Go deep on [Module] today”

Strategic identity titles name the difference clearly. This is the primary driver of the two-option proposal — not situational. Always breadth vs. depth.

Phone: swipe between cards with paginator dots. Desktop: side by side.

“Why this plan?” panel

- Collapsed by default, always
- Tap “Why this plan? ↓” to reveal
- Color-coded dot per block, one concise reason per module
- Surfaces pace deficit language when relevant (see Section 2)

Feasibility banner

Sits **above the session stack** on the proposal screen when any goal is behind pace. Same design as Practice Sessions home banner (see Section 3) — most urgent goal leads, rest collapsible.

Time adjustment — tappable on proposal screen

The time allocation (“45 min”) shown in the proposal header is **directly tappable**. Tapping opens an inline time picker (same presets as input Question 1: 15 / 30 / 45 / 60 / 90 / custom). Changing the time regenerates the proposal instantly — no re-entering the input flow.

“+ Add block” picker

At the bottom of the session stack, a subtle “+ Add block” affordance. Tapping opens a three-option picker:

1. Go deeper on something already in this session

- Existing session blocks shown as selectable cards
- User taps one, declares how much more time

- Adds a second block for that module/item ("Practice Mirror for another 30 min")

2. Next priority (algorithm suggestion)

- Single recommended item: "Based on your goals, the next best use of your time is —"
- One-tap add

3. Pick your own

- Browse by module
- Algorithm surfaces next-priority suggestions within each module
- User can override completely — full agency

Covers three real scenarios: in-flow and want to stay there (option 1), want to stay optimal (option 2), know exactly what you want (option 3).

5. Phase 2 updated step list

Phase 2 now has 7 steps (was 6):

1.  spacingState population (all 8 modules)
2.  Coverage metric vocabulary + GoalCreationFlow wiring
3.  moduleItemCounts helper
4.  Coverage + accuracy progress helpers
5.  YearlyAnchorFlow UI (in progress — 5a, 5b, 5c complete; 5d–5g remaining)
6. Goals home redesign (by-timeframe + by-module views, expanded goal rows, YearlyAnchorFlow integration)
7. Goal feasibility checking — `getGoalFeasibility()` helper + Goals home feasibility surface (NEW)

Practice Sessions home feasibility banner deferred to Phase 3.

6. Items NOT yet designed (still needed before Phase 3 build)

The following Phase 3 UI design questions were identified but not yet resolved. Design these before Phase 3 build starts:

1. **Block visualization**  Complete — see Section 4
2. **Input questionnaire UX** — one screen vs. stepped flow? Bottom sheet or full screen?
How does the time input relate to the tappable time on the proposal screen?
3. **Cold-start experience** — what does first session generation look like with goals but minimal spacingState history?
4. **Song Practice Timer** — still fully undesigned. Needs
SONG_PRACTICE_TIMER_DESIGN.md before Phase 3 touches songs.